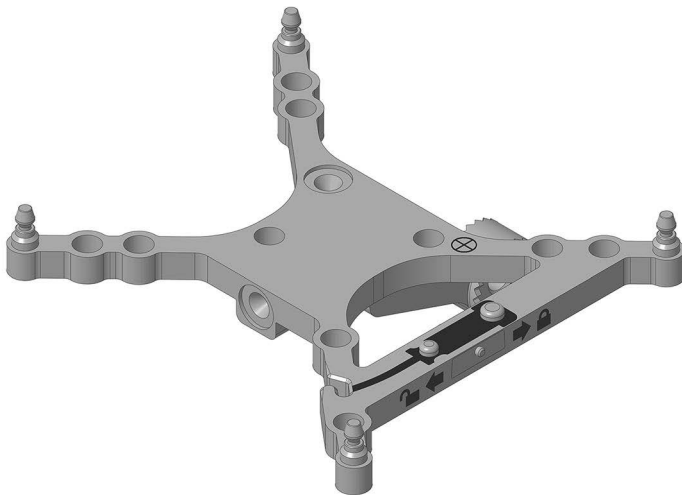


**EN**

Hybrid Reference Frame

(Component not shown to scale)

1 Hybrid Reference Frame (9736511)

Device description

The Hybrid Reference Frame is a reusable reference frame that is an accessory to your Stealth™ system. The Hybrid Reference Frame is used for localization during spine surgeries. It can be used in all spine procedures where automatic registration can be used, including with intraoperative scanners that can be used for Spine StealthAIR™ Registration. The Hybrid Reference Frame also features multiple verification divots for navigated instrument verification.

The Hybrid Reference Frame contains a mating component for the EM Spine Reference Frame Tracker to attach to during spine procedures that incorporate the Segmental Tracking Feature.

Intended use

The Hybrid Reference Frame is intended to enable localization of patient anatomy during navigated procedures with Medtronic computer-assisted surgery systems.

Indications for use

The Stealth AXIS™ Surgical System is intended for precise positioning of surgical instruments and as an aid for precisely locating anatomical structures in open, minimally invasive, and percutaneous procedures. The Stealth AXIS™ Surgical System is indicated for medical conditions in which the use of stereotactic surgery may be appropriate, and where reference to a rigid anatomical structure, such as the skull or vertebra, can be identified relative to a CT or MR based model, fluoroscopy images, or digitized landmarks of the anatomy.

Contraindications

None.

Material composition

For materials of concern information such as REACH, CA Prop 65, or other product stewardship programs, go to www.medtronic.com/productstewardship.

Compatibility

The Hybrid Reference Frame is compatible with spine referencing instruments that have the correct mating starburst adaptor. For example, spinous process clamps or percutaneous pins. Refer to the instructions for use

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that accompany these spine referencing fixation devices for information about attaching the Hybrid Reference Frame to the patient.

System level manuals

This device can be used with Medtronic computer-assisted surgery systems. See the system level manuals and clinical application instructions for use for descriptions of patient groups, intended users, potential clinical benefits, side effects, and potential complications. This device is an accessory to your system.

See the system level manuals and the spine clinical application instructions for use for additional instructions, warnings, and precautions.

Content of this document

This document contains use instructions for the Hybrid Reference Frame.

Warnings

Warning: Before first use and for reuse, clean and sterilize the device according to the instructions in this document. The device is designed for reuse and is supplied non-sterile. Use of a non-sterilized device could result in patient injury or death.

Warning: Do not open sterile-barrier packaging or containers until surgical use. At the time of surgical use, inspect barrier for breach. If the sterile barrier was breached prior to use abandon use of the device.

Warning: Only use Medtronic supported sterile spheres on the Hybrid Reference Frame to prevent inaccurate navigation information, which could result in serious injury or death.

Warning: Do not bump or reposition the patient reference frame after registration. Movement of the patient reference frame relative to the patient anatomy might result in inaccurate navigation and could result in serious injury or death. If the patient reference frame moves in relation to the patient anatomy at any time after registration, verify accuracy by navigating on known anatomical landmarks. You must re-register if there is inaccurate navigation. See the *EM Spine Reference Frame Tracker* instructions for use regarding bump or reposition considerations when using segmental tracking.

Warning: Do not move the EM Spine Reference Frame Tracker after patient registration. Movement of the EM Spine Reference Frame Tracker relative to its initial position on the Hybrid Reference Frame might result in inaccurate navigation and could result in serious injury or death. If you remove and replace the EM Spine Reference Frame Tracker at any time after registration, verify accuracy by navigating on known anatomical landmarks. You must re-register if there is inaccurate navigation. See the *EM Spine Reference Frame Tracker* instructions for use regarding reference frame placement considerations when using segmental tracking.

Warning: Abort use of the system if system navigation seems inaccurate and recommended steps to restore accuracy are not successful. Continued use of the system with inaccurate navigation could result in serious injury or death.

Warning: Abandon use of the device if the device sterility is compromised during the procedure. Use of a non-sterilized device could result in serious injury or death.

Warning: Remove single-use optical spheres before cleaning and sterilization. Failure to heed this warning will prevent the device from being effectively cleaned and sterilized and could result in serious injury or death.

Precautions

Caution: Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.

Use

Follow these instructions to use the Hybrid Reference Frame.

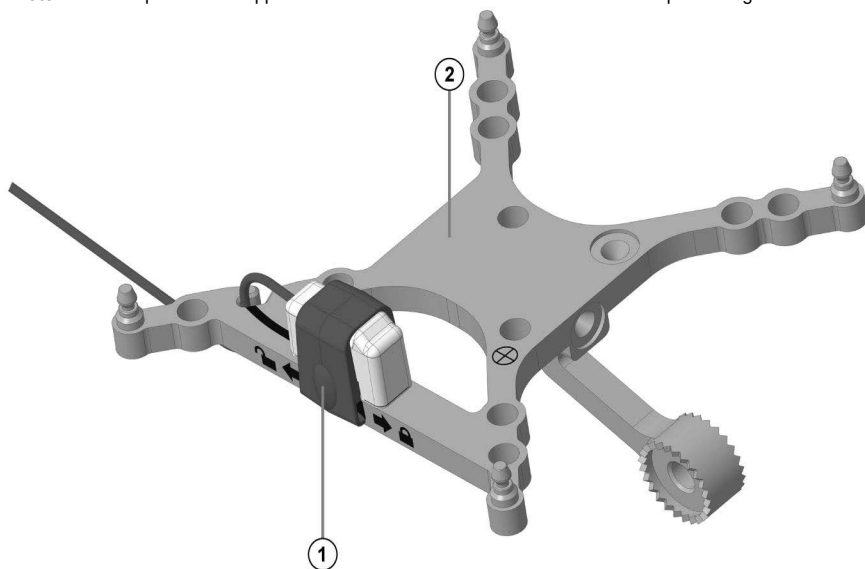
Note: Refer to the instructions for use that accompany the O-arm™ imaging system for additional information about using the Hybrid Reference Frame for O-arm™ scans and additional features.

1. Snap sterile spheres onto the Hybrid Reference Frame. Push each sphere onto the stem until it “clicks” into place. Make sure that the sphere is firmly seated on the stem.
2. If you are using the Hybrid Reference Frame in a spine procedure with segmental tracking, attach the EM Spine Reference Frame Tracker to the Hybrid Reference Frame. Ensure proper orientation and connection of the EM Spine Reference Frame Tracker to the Hybrid Reference Frame.

Warning: Do not apply excessive force when attaching the EM Spine Reference Frame Tracker to the cable hook of the Hybrid Reference Frame. Applying excessive force may damage the cable.

Note: See the *EM Spine Reference Frame Tracker* instructions for use for information about how to attach the EM Spine Reference Frame Tracker.

Note: See the Spine clinical application instructions for use for information about patient registration.



- 1 EM Spine Reference Frame Tracker
2 Hybrid Reference Frame

3. Attach the Hybrid Reference Frame to the patient using any referencing option that is compatible with the Hybrid Reference Frame.
 4. For proper placement of the Hybrid Reference Frame, consider the following:
 - The sterile spheres must be visible to the camera on the camera cart.
 - Position the Hybrid Reference Frame so that it allows access to desired trajectories and anatomical areas of interest during the procedure.
 - If using the Hybrid Reference Frame for StealthAIR™ Spine registration, place the frame within 300 mm (12.0 in) of the surgical area of interest to ensure anatomy is within the navigable volume.
 - See the *EM Spine Reference Frame Tracker* instructions for use for information about how to place the tracker relative to the emitter when using the EM Spine Reference Frame Tracker for segmental tracking.
- Warning:** Ensure that the reference frame is rigid with respect to anatomy. Movement of the patient reference frame relative to the patient anatomy after patient registration will result in inaccurate navigation and could result in serious injury or death.
5. In the clinical application **Equipment** task, add the Hybrid Reference Frame.
 6. In the clinical application **Instruments** task, add the Hybrid Reference Frame to the instruments list.
 7. Tap the **Tracking View** button to open the **Tracking View** and then confirm that the Hybrid Reference Frame is in the camera's field of view.
 8. If using the Hybrid Reference Frame in a segmental tracking procedure, remove the EM Spine Reference Frame Tracker from the Hybrid Reference Frame at the end of the procedure.
 - a. Slide the clip toward the cable connection to release the latch.
 - b. Pull the EM Spine Reference Frame Tracker off the two posts to remove it from the frame.
 - c. Dispose of the single-use EM Spine Reference Frame Tracker.

Device processing - cleaning, inspection and sterilization

These instructions are intended to assist hospital personnel in safely handling and effectively processing devices supplied non-sterile by Medtronic. The specified devices must be cleaned, inspected, packaged, and sterilized prior to initial use and each subsequent use.

General warnings and precautions	Warning: Use CE-marked processing equipment that has been properly validated, calibrated, and maintained. Warning: To prevent cross-contamination, always handle, transport, clean, sterilize, and package devices contacting the central nervous system separately from other devices. Warning: Ensure that all processing steps are properly and completely executed before releasing the device for use. Warning: Do not use the device if there are any signs of damage, deterioration, or deformation. Discontinue use of the device immediately if it becomes damaged during the procedure.
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Limitations on reprocessing	The useful life of this device depends on a variety of factors including method and duration of each use, as well as handling between uses. End of useful life is normally determined by wear and damage due to use. Careful inspection and functional testing prior to use is the best method for determining if the device is at the end of its useful life. Refer to the 'Maintenance, inspection, and testing' section below for additional instruction. Warning: Prior to use, examine the device components for damage, deterioration, deformation, and abuse. Do not attempt to use any device that appears to be bent or otherwise damaged. Abandon the use of any device damaged during the procedure. Refer to the 'Maintenance, inspection, and testing' instructions in this document to determine if the device is suitable for use. If defects are found, do not use the product, and contact Medtronic immediately. Warning: Do not reuse or reprocess a device where prion contamination is suspected or confirmed. Incinerate the potentially contaminated device according to national regulations. Warning: At the end of its useful life, dispose of the device in accordance with national regulations.
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Instructions

Initial treatment at the point-of-use	1. Processing begins at the point of use. Do not allow blood, debris, or bodily fluids to dry on the device. 2. Remove gross contamination from the device using non-linting or non-shedding wipes or cloths. 3. If the device cannot be reprocessed immediately, keep the device moist during transport by placing the device in a tray of utility water, if available, or by covering the device with a damp towel. 4. To prolong the life of the device, process immediately after use. Contain and transport the device in a closed, puncture proof container per OSHA or local regulations. Soiled devices should be transported separately from non-contaminated devices to avoid contamination of personnel and surroundings. Warning: Use personal protective equipment, as necessary, when handling, processing, and disposing the device. Universal precautions for handling devices after use should be observed by all hospital personnel according to OSHA Standard 29 CFR 1910.1030, Occupational Exposure to Bloodborne Pathogens, or local regulations. Warning: Devices should be cleaned within 30 minutes of use to limit fixation of contaminants.
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Preparation before cleaning	Warning: Incorrect removal of sterile spheres may damage the device. 1. Remove single-use spheres and dispose of them in accordance with national regulations. No special tools are required; grasp the sphere at its base and pull. 2. Thoroughly rinse the device using cold running utility water (10-22°C [50-72°F]) for at least 30 seconds and until all visible soil is removed. While rinsing, actuate any movable components through their full range of motion (if applicable) and use a soft-bristled nylon brush (such as Key Surgical, part number N-2100) and clean non-linting wipe to aid in soil removal. Special attention should be given to any lumens, blind holes, mated surfaces, crevices, cavities and other features that may present a challenge to cleaning. 3. Carefully inspect the device to ensure that all visible soil has been removed. If any soil remains on the device, continue rinsing and brushing the device until it is visibly clean. Warning: Do not use metal brushes, scouring pads, or abrasive cleaners as this may damage the device. Caution: Do not use processing agents that may cause corrosion (such as saline or cleaning agents that contain aldehyde, active chlorine, bromine, or iodide).																																		
Cleaning: auto-mated	1. Transfer the device to a washer-disinfector complying with ISO 15883. The device should be positioned such that all surfaces are easily accessible, device lumens, blind holes, and cavities are facing down, and hinges are open (to allow for draining). 2. Select the appropriate cycle parameters based on detergent type (that is, enzymatic or alkaline cleaning cycle). 3. Upon completion of the automated wash cycle, carefully inspect all parts of the device for visible soil. If any soil is observed, repeat the cleaning process. Warning: Use the rigid container and instrument tray for sterilization only. Instruments must be cleaned separate from the container or tray. Warning: Do not place heavy instruments on top of delicate devices to prevent damage to the devices. Note: Cleaner and detergent manufacturers provide concentrated solutions, which must be diluted per the manufacturer's instructions for cleaning of medical devices. <table><tr><th colspan="4">Enzymatic cleaning cycle</th></tr><tr><th>Phase</th><th>Recirculation/ Soak Time</th><th>Water Temperature</th><th>Detergent Type</th></tr><tr><td>Prewash</td><td>2 minutes (02:00)</td><td>Cold utility water 10-22°C (50-72°F)</td><td>N/A</td></tr><tr><td>Enzyme wash</td><td>2 minutes (02:00)</td><td>Hot utility water 43-55°C (109-131°F)</td><td>Neutral pH enzymatic cleaner</td></tr><tr><td>Wash</td><td>2 minutes (02:00)</td><td>Hot utility water 66°C (151°F) (setpoint)</td><td>Neutral pH detergent</td></tr><tr><td>Rinse</td><td>15 seconds (00:15)</td><td>Hot utility water 43-60°C (109-140°F)</td><td>N/A</td></tr><tr><td>Thermal disinfection</td><td>1 minute (01:00)**</td><td>Hot critical water ≥ 90°C (194°F)</td><td>N/A</td></tr><tr><td>Dry</td><td>30 minutes (30:00)</td><td>100°C (212°F) (setpoint)</td><td>N/A</td></tr></table> Note: Enzymatic cycle parameters were validated using STERIS Valsure™ Enzymatic Cleaner and Neutral Detergent. **Thermal disinfection has been validated for 1 minute at 90°C (A₀ ≥ 600) and for 5 minutes at 90°C (A₀ ≥ 3000).			Enzymatic cleaning cycle				Phase	Recirculation/ Soak Time	Water Temperature	Detergent Type	Prewash	2 minutes (02:00)	Cold utility water 10-22°C (50-72°F)	N/A	Enzyme wash	2 minutes (02:00)	Hot utility water 43-55°C (109-131°F)	Neutral pH enzymatic cleaner	Wash	2 minutes (02:00)	Hot utility water 66°C (151°F) (setpoint)	Neutral pH detergent	Rinse	15 seconds (00:15)	Hot utility water 43-60°C (109-140°F)	N/A	Thermal disinfection	1 minute (01:00)**	Hot critical water ≥ 90°C (194°F)	N/A	Dry	30 minutes (30:00)	100°C (212°F) (setpoint)	N/A
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Dry	30 minutes (30:00)	100°C (212°F) (setpoint)	N/A																																

Alkaline cleaning cycle			
Phase	Recirculation/ Soak Time	Water Temperature	Detergent Type
Prewash	2 minutes (02:00)	Cold utility water 10-22°C (50-72°F)	N/A
Alkaline wash	5 minutes (05:00)	Hot utility water 43°C (109°F) (setpoint)	Alkaline detergent
Rinse	1 minute (01:00)	Hot utility water 43-60°C (109-140°F)	N/A
Thermal disinfection	1 minute (01:00)**	Hot critical water ≥ 90°C (194°F)	N/A
Dry	30 minutes (30:00)	100°C (212°F) (setpoint)	N/A
Note: Alkaline cycle parameters were validated using Dr. Weigert Neodisher™ Mediclean Forte.			
**Thermal disinfection has been validated for 1 minute at 90°C (A ₀ ≥ 600) and for 5 minutes at 90°C (A ₀ ≥ 3000).			
Drying	If a dry phase is not included in the washer-disinfector cycle: • Dry the device thoroughly inside and out to prevent rusting and malfunction. • Dry the device with a clean, soft, lint-free towel. • Special attention should be given to threads, hinges, crevices, cavities, and other areas where fluid can accumulate. • Open and close the device (if applicable) so that all areas are accessible. • Dry lumens, blind holes, and mated surfaces using instrument grade compressed air.		
Maintenance, inspection, and testing	1. Visually inspect the device under good light conditions for completeness, damage, and excessive wear. Consideration should be given to the following: - All device components should be accounted for. - Laser etchings, engravings, and other markings should be legible. - No chips, cracks, or sharp-edged gouges should be present. - No bending, wearing, warping or pitting should be present. - No discoloration, rust, or corrosion should be present. 2. If any of the criteria listed above is not met, then the device is at the end of its useful life and should be disposed of according to national regulations. Warning: Do not process or use any device that has obvious damage or corrosion.		
Packaging	The device should be packaged for sterilization by placement in a Medtronic-branded sterilization tray within a Medtronic-branded reusable rigid sterilization container (9736633, 9736632, 9736655, or 9736654) with disposable filters. Warning: Do not place additional devices inside the sterilization tray as this may impact sterilization efficacy and dry time. Warning: Do not wrap reusable rigid sterilization containers. The device may not be sterilized to an adequate sterility assurance level. Note: The user facility is responsible for using only accessories (such as chemical indicators, and biological indicators) that have been approved by the Food and Drug Administration for medical facilities inside the USA and its territories or conform to EN ISO 11607 for medical facilities outside the USA and its territories.		
Sterilization	1. Load the packaged device into the steam sterilizer according to the sterilizer's operating manual. 2. Program the steam sterilizer to deliver one of the following validated pre-vacuum (dynamic air removal) steam (moist heat) sterilization cycles:		
Cycle Type	Temperature	Exposure Time	Minimum Dry time*
For Healthcare Facilities in any geography			
Pre-vacuum	132°C (270°F)	4 minutes (04:00)	30 minutes (30:00)
Pre-vacuum	135°C (275°F)	3 minutes (03:00)	30 minutes (30:00)
For Healthcare Facilities outside the USA and its territories			
Pre-vacuum	134°C (273°F)	3 minutes (03:00)	30 minutes (30:00)
Pre-vacuum	134°C (273°F)	18 minutes (18:00)	30 minutes (30:00)

	*Minimum dry times were validated using sterilizers with vacuum drying capabilities. Additional time may be required to dry the device under ambient atmospheric pressure or variable environmental conditions. Warning: Do not sterilize and supply for surgical use any device that is not visibly clean and free of particulates. If particulates are present, repeat reprocessing, starting with the Preparation before cleaning step. Warning: Device may not be sterilized to an adequate sterility assurance level without prior cleaning. Warning: Do not stack containers or trays during sterilization as this may limit steam penetration and prevent effective sterilization. Note: Steam for sterilization should be generated from water that has been treated to remove total dissolved solids, filtered to remove contaminants and water droplets, and supplied via piping without dead legs or other stagnant zones where contamination might collect. Steam saturation should be greater than 97%.
Storage	1. The device should remain in its sterilization packaging and be stored in a dry and clean location under ambient temperature conditions. 2. Consideration should be given to the sterile barrier system, storage manner, and environmental conditions when determining the maximum amount of time permitted between processing and use.
Additional information	• Definitions of and performance qualification levels for utility water and critical water are described in ANSI/AAMI ST108. • In the event of an alleged product failure, it is important to return the device to Medtronic for investigation; however, Medtronic is only able to accept devices that are visibly clean and free of contaminants.

Medtronic has validated the processes provided in these instructions in accordance with ANSI/AAMI ST98, ISO 17665, AAMI TIR12 and ISO 17664 and demonstrated that they are capable of yielding product that meets quality, safety, and efficacy requirements. It remains the responsibility of the healthcare facility to ensure that processing, as performed using the equipment, materials, and operators at their facility, consistently achieves the desired result. This requires verification or validation or both, and routine monitoring of the process. Alternative methods of processing outside the scope of these instructions may be suitable; however, they must be validated by the end user. In the event of conflicting national cleaning or sterilization requirements, those shall prevail over Medtronic's recommendations.

Sterile spheres

During a surgical procedure, a sterile sphere may become coated with fluid or debris, hindering its ability to reflect light. If this occurs, wipe away the debris with a soft, sterile cloth (the cloth may be wet, but make sure to dry the sphere thoroughly). Keep extra sterile spheres available as replacements during a case.

Passive sterile spheres are presterilized, single-use components. Do not resterilize or reuse.

Warning: Do not reprocess, resterilize, or reuse any single-use device. Attempts to reprocess or resterilize a single-use device may be ineffective and may compromise its structural integrity. Any reuse creates a risk of contamination which could result in patient injury, illness, or death.

Disposal

Dispose of contaminated devices according to your facility's biohazard disposal procedure and local and national regulations.

Contact information

If a serious incident occurs in relation to the use of this device, report it to Medtronic Navigation. If a serious incident occurs in the European Union, also report it to the competent authority in the Member State where the incident occurred.

Assistance

If you have questions about this device, contact your local Medtronic Navigation representative or call Medtronic Navigation, Inc. at 1 800 595 9709 or +1 720 890 3160.

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This product may be protected by US Patents listed at: medtronic.com/patents.

Symbol definitions

The following symbols may appear on the device or the device packaging:

Rx Only



Caution: Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.

Consult electronic instructions for use.



The EM Spine Reference frame tracker is not locked in place.



The EM Spine Reference frame tracker is locked in place.

R_x Only

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